Implementation reports (area / timing / power)

All synthesis and place-and-route reports for the SNN packet classifier, in one place, with a comparison to the literature. Numbers are copied from the actual tool reports (paths given per section). The source `.rpt`/`.summary` files are committed in the repo.

Quick answer — does each flow report area / timing / power?

flow	area	timing	power
sky130 synthesis (Yosys)	✓ cell area	✗ (STA not run in the open synth script)	✗
sky130 place & route (OpenROAD)	✓ die + cell area	✓ multi-corner	✓ (vectorless estimate)
FPGA Quartus (Cyclone V)	✓ **resources (ALM/reg/RAM/DSP)**	✓ **Fmax / slack**	✗ **PowerPlay not run**
FPGA Tang Nano 9K (nextpnr)	✓ LUT/DFF/BSRAM	✓ Fmax	✗ (no open power tool for Gowin)
Cadence 90 nm (Genus/Innovus)	✓	✓	✓

So the **FPGA reports give area and timing, but not power.** FPGA power needs a separate PowerPlay run (`quartus_pow`) with a switching-activity file (`.vcd`/`.saif`); that hasn't been run. The only measured **power** number so far is the sky130 place-and-route estimate (125 mW).

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1. ASIC — SkyWater sky130 (open-source)

1a. Synthesis (Yosys + ABC), cell area

Reports: `asic/sky130/out/*_area.rpt` (logs: `asic/sky130/out/*_yosys.log`, summaries `asic/sky130/out_*.txt`). Timing reports (`*_timing.rpt`) are empty — the open synth script (`asic/sky130/synth.sh`) does synthesis only (OpenSTA isn't installed locally); real timing comes from OpenROAD below.

module	cells	cell area (μm²)	cell area (mm²)
`snn_core` (16+2 LIF neurons + W1 memory)	—	393,195	0.39
`snn_classifier` (parser + encoder + core)	—	433,832	0.43
`snn_pkt_filter_asic` (full chip, + 2 KiB buffer)	69,330	1,148,869	**1.15**

Sequential share of the full chip: **29,166 flip-flops = 858,000 µm² ≈ 75 %** of the cell area (`edfxtp_1` ×27,366 + `dfxtp_1` ×1,800). Most of that is the two flop-based memories (packet buffer + W1 weights); multipliers: **0**.

1b. Place & route (OpenROAD via LibreLane) — area, timing, power

Full results on the GitHub release `gds-snn_classifier` (`report.md`, `metrics.json`). Block: `snn_classifier`.

metric	**value**
die area	1.28 mm² (1.13 × 1.14 mm)
std-cell area	0.707 mm², 72,032 cells
timing	≈ **99 MHz** typical (setup WNS −0.077 ns); ≈ 136 MHz fast corner; ≈ 51 MHz slow corner
power	**125 mW** @ 100 MHz, typical corner (vectorless estimate): 82.9 mW internal + 42.5 mW switching + ~0 leakage
DRC / LVS	0 / 0
multipliers	0

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2. FPGA — Cyclone V (Chameleon96, full HPS + SNN system)

Reports: `fpga/cv96_snn/output_files/led_blink_linux.fit.summary` (area), `*.sta.summary` (timing), `*.fit.rpt` / `*.sta.rpt` (full). Tool: Quartus Prime 20.1 Lite.

2a. Area / resources (`.fit.summary`)

resource	**used**	**available**	**%**
Logic (ALMs)	2,378	41,910	6 %
Registers	3,080	—	—
Block memory bits	9,776	5,662,720	< 1 %
RAM blocks	8	553	1 %
DSP blocks	**0**	112	**0 %**
I/O pins	51	205	25 %

(This is the whole system: the HPS interface + the SNN accelerator; the SNN alone is a small part.)

2b. Timing (`.sta.summary`, Slow 1100 mV 100 °C)

clock	**setup slack**	**hold slack**
`h2f_user0_clk` (the SNN clock, 100 MHz)	**+0.975 ns**	+0.243 ns
HPS SDRAM write clock	+2.407 ns	+0.164 ns

Positive slack everywhere → **meets 100 MHz**; the SNN clock's restricted Fmax is **110.8 MHz**.

2c. Power

Not reported. PowerPlay (`quartus_pow`) was not run, so there is no FPGA power number. To get one: run `quartus_pow` with a `.vcd`/`.saif` from a gate-level sim.

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3. FPGA — Cyclone V (in-line filter, standalone fit_check)

Report: `fpga/fit_check/output_files/snn_fw_fit.fit.summary`. This fits just the in-line filter (bigger 4 KiB buffer) on its own, to size the datapath alone.

resource	used	%
Logic (ALMs)	5,871	14 %
Registers	10,336	—
RAM blocks	14	3 %
DSP blocks	0	0 %

Standalone Fmax ≈ 111.6 MHz (from the STA report).

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4. FPGA — Tang Nano 9K (Gowin GW1NR-9, open flow)

Report: `fpga/tang9k/nextpnr.log` (Yosys `synth_gowin` → nextpnr-himbaechel → gowin_pack).

resource	used	available	%
LUT4	7,035	8,640	81 %
DFF	2,657	6,480	41 %
BSRAM	6	26	23 %
MUX2_LUT5	99	4,320	2 %
I/O	12	276	4 %
DSP	**0**	—	—

Timing: Max frequency **67.3 MHz** vs the 27 MHz board clock (2.5× margin). **Power:** no open-source power tool for Gowin, so not reported.

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5. Cadence 90 nm (Genus / Conformal / Innovus)

Run on the university server (`asic/README.md`, `asic/scripts/run_flow.sh`). Area, timing **and power** come out of Genus (`report_area`, `report_timing`, `report_power` → `asic/out/<design>/syn_*/reports/`) and Innovus (post-route). These reports are tracked in the repo once committed from the server — **being finalised**.

6. Summary across all targets

target	flow	area	clock	power
sky130	OpenROAD P&R	1.28 mm² die, 72,032 cells	≈ 99 MHz (tt)	125 mW
sky130	Yosys synth (full chip)	1.15 mm², 69,330 cells	n/a	n/a
Cyclone V	Quartus (HPS + SNN)	2,378 ALMs (6 %), 0 DSP	110.8 MHz (meets 100)	not measured
Cyclone V	Quartus (filter only)	5,871 ALMs (14 %), 0 DSP	111.6 MHz	not measured
Tang Nano 9K	nextpnr	7,035 LUT4 (81 %), 0 DSP	67.3 MHz (uses 27)	not measured
Cadence 90 nm	Genus/Innovus	pending	8 ns target	pending

Zero multipliers / DSP blocks on every FPGA target — the point of using a spiking network.

7. Comparison with the literature

How these numbers sit against published SNN / in-network-ML hardware. Most prior SNN intrusion-detection work runs *off* the data path (software or a neuromorphic chip fed by software); in-network ML (Taurus) uses *non-spiking* models in a switch; generic FPGA SNN accelerators target images. Ours is an SNN fused into the NIC datapath, at line rate, with no multipliers.

work	platform	multipliers	area / power note	vs ours
Alom & Taha [C1]	TrueNorth chip	—	neuromorphic ASIC	SNN IDS 90.1 %, but off the data path (software feeds it)
Zhou & Li [C2]	software / sim	—	temporal (single-spike) coding	SNN IDS algorithm; no line-rate hardware
Taurus [C3]	switch ASIC	many	per-packet ML in-switch	non-spiking (DNN/SVM); not at the NIC→host boundary
Plagwitz et al. [J2]	FPGA	SNN: none	measures SNN vs CNN area & power	shows SNN beats CNN on power **only when sparse** — matches our event-driven, 0-

work	platform	multipliers	area / power note	vs ours
FPGA SNN accel. [C4]	FPGA	none	image benchmarks	DSP design not fused with a NIC
Na & Kim [J3]	ASIC	—	energy-efficient neuromorphic NoC	edge-AI processor, not packet-oriented
Ours	ASIC + FPGA NIC	**0**	**1.28 mm² / 125 mW (sky130); 6 % ALM, 0 DSP (FPGA)**	**SNN fused in the NIC datapath, line rate, 0 DSP**

Our FPGA result (6 % of a Cyclone V, **0 DSP blocks**, 110 MHz) and the sky130 figures (1.28 mm², 125 mW) are consistent with the literature's finding that SNNs win on hardware cost **when they are sparse and multiplier-free** [J2]. Accuracy is still on synthetic data; published SNN-IDS on real data is ≈ 90 % [C1] (see `docs/training.md`).

References

Journal - **[J1]** Neftci, Mostafa, Zenke, "Surrogate Gradient Learning in Spiking Neural Networks," *IEEE Signal Processing Magazine*, 2019. - **[J2]** Plagwitz, Hannig, Teich, Keszöcze, "To Spike or Not to Spike? A Quantitative Comparison of SNN and CNN FPGA Implementations," *ACM Trans. Embedded Computing Systems*, 2023. - **[J3]** Na, Kim, "An Energy-Efficient Neuromorphic Processor Using Unified Refractory Control-Based NoC for Edge AI," *Electronics (MDPI)*, 2025. - **[J4]** "Algorithm–hardware co-design of neuromorphic networks with dual memory pathways," *Nature Machine Intelligence*, 2026.

Conference - **[C1]** Alom, Taha, "Network Intrusion Detection for Cyber Security on Neuromorphic Computing System," *IEEE IJCNN*, 2017. - **[C2]** Zhou, Li, "Spiking Neural Networks with Single-Spike Temporal-Coded Neurons for Network Intrusion Detection," *IEEE ICPR*, 2020. - **[C3]** Swamy et al., "Taurus: A Data Plane Architecture for Per-Packet ML," *ACM ASPLOS*, 2022. - **[C4]** "Efficient Hardware Acceleration of Spiking Neural Networks using FPGA," *IEEE VTC2024-Spring*, 2024.

Datasets: Moustafa & Slay, UNSW-NB15 (MilCIS 2015); Sharafaldin et al., CIC-IDS2017 (ICISSP 2018).

Report locations: sky130 `asic/sky130/out/`; Cyclone V `fpga/cv96_snn/output_files/` and `fpga/fit_check/output_files/`; Tang `fpga/tang9k/nextpnr.log`; sky130 P&R the `gds-snn_classifier` GitHub release; Cadence `asic/out/<design>/.../reports/`.